Engineering Economy

Chapter 7: **Depreciation and Income Taxes**

The objective of Chapter 7 is to explain **how depreciation affects income taxes**, and **how income taxes affect economic decision making**.

Chapter 7
7.1. Introduction

Income taxes usually represent a significant cash outflow.

In this chapter we describe how after tax liabilities and after-tax cash flows result in the *after-tax cash flow* (ATCF) *procedure*.

Depreciation is an important element in finding after-tax cash flows.

Chapter 7
7.2. Depreciation Concept and Terminology

Depreciation is the decrease in value of physical properties with the passage of time.

- It is an accounting concept, a *non-cash* cost, that establishes an annual deduction against before-tax income.
- It is intended to approximate the yearly fraction of an asset's value used in the production of income.

Property is depreciable if ;

- it is used in business or held to produce income.
- it has a determinable useful life, longer than one year.
- it is something that wears out, decays, gets used up, becomes obsolete, or loses value from natural causes.
- it is not inventory, stock in trade, or investment property.

Chapter 7
7.2. Depreciation Concept and Terminology

Depreciable property is

- tangible (can be seen or touched; personal or real) or intangible (such as copyrights, patents, or franchises).
- depreciated, according to a depreciation schedule, when it is put in service (when it is ready and available for its specific use).

Chapter 7
7.3. The Classical Depreciation Methods

There are different methods to calculate the value loss of physical properties

- Straight-Line (SL) Method
- Decline-balance Method (DB)
- Units of Production Method

Chapter 7
7.3. The Classical Depreciation Methods

7.3.1. Straight line (SL):
constant amount of depreciation each year over the depreciable life of the asset.

$$d_k = \frac{B - SV_N}{N}$$

- N = depreciable life
- B = cost basis
- d_k = depreciaton in k
- BV_k = book value at end of k
- SV_N = salvage value

Chapter 7
7.3. The Classical Depreciation Methods

Pause and solve

Acme purchased a coordinate measurement machine (CMM). The cost basis is $120,000 and it has a seven year depreciable life. Acme estimates a salvage value of $22,000 at the end of seven years. Determine the annual depreciation amounts using SL depreciation. Tabulate the annual depreciation amounts and book value of the CMM at the end of each year.

Solution

B = $120,000
SV = $22,000
N = 7 years

$$d_k = \frac{120,000 - 22,000}{7} = \$14,000$$

Year	Depreciation	Ending Book Value
0	-	$120,000
1	$14,000	106,000
2	$14,000	$92,000
3	$14,000	$78,000
4	$14,000	$64,000
5	$14,000	$50,000
6	$14,000	$36,000
7	$14,000	$22,000

Chapter 7
7.3. The Classical Depreciation Methods

EXAMPLE 7-1 **SL Depreciation**



A laser surgical tool has a cost basis of $200,000 and a five-year depreciable life. The estimated SV of the laser is $20,000 at the end of five years. Determine the annual depreciation amounts using the SL method. Tabulate the annual depreciation amounts and the book value of the laser at the end of each year.

Solution

The depreciation amount, cumulative depreciation, and BV for each year are obtained by applying Equations (7-2), (7-3), and (7-4). Sample calculations for year three are as follows:

$$d_3 = \frac{\$200{,}000 - \$20{,}000}{5} = \$36{,}000$$

$$d_3^* = 3\left(\frac{\$200{,}000 - \$20{,}000}{5}\right) = \$108{,}000$$

$$\mathrm{BV}_3 = \$200{,}000 - \$108{,}000 = \$92{,}000$$

Chapter 7
7.3. The Classical Depreciation Methods

EXAMPLE 7-1 **SL Depreciation**

The depreciation and BV amounts for each year are shown in the following table.

EOY, k	d_k	BV_k
0	—	$200,000
1	$36,000	$164,000
2	$36,000	$128,000
3	$36,000	$92,000
4	$36,000	$56,000
5	$36,000	$20,000

Note that the BV at the end of the depreciable life is equal to the SV used to calculate the yearly depreciation amount.

7.3.2. Declining-balance (DB):

a constant-percentage of the remaining BV is depreciated each year.

$$d_k = B(1-R)^{k-1}(R) \qquad BV_k = B(1-R)^k$$

The constant percentage is determined by *R*, where *R = 2/N* when 200% declining balance is being used, *R = 1.5/N* when 150% declining balance is being used.

Chapter 7
7.3. The Classical Depreciation Methods

EXAMPLE 7-2 **DB Depreciation**



A new electric saw for cutting small pieces of lumber in a furniture manufacturing plant has a cost basis of $4,000 and a 10-year depreciable life. The estimated SV of the saw is zero at the end of 10 years. Use the DB method to calculate the annual depreciation amounts when

(a) $R = 2/N$ (200% DB method)

(b) $R = 1.5/N$ (150% DB method).

Tabulate the annual depreciation amount and BV for each year.

Solution

Annual depreciation, cumulative depreciation, and BV are determined by using Equations (7-6), (7-7), and (7-8), respectively. Sample calculations for year six are as follows:

(a)

$$R = 2/10 = 0.2,$$
$$d_6 = \$4{,}000(1 - 0.2)^5(0.2) = \$262.14,$$
$$d_6^* = \$4{,}000[1 - (1 - 0.2)^6] = \$2{,}951.42,$$
$$\text{BV}_6 = \$4{,}000(1 - 0.2)^6 = \$1{,}048.58.$$

(b)

$$R = 1.5/10 = 0.15,$$
$$d_6 = \$4{,}000(1 - 0.15)^5(0.15) = \$266.22,$$

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7.3. The Classical Depreciation Methods

EXAMPLE 7-2 **DB Depreciation**

$$d_6^* = \$4{,}000[1 - (1 - 0.15)^6] = \$2{,}491.40,$$

$$\text{BV}_6 = \$4{,}000(1 - 0.15)^6 = \$1{,}508.60.$$

The depreciation and BV amounts for each year, when $R = 2/N = 0.2$, are shown in the following table:

200% DB Method Only		
EOY, k	d_k	BV_k
0	—	\$4,000
1	\$800	3,200
2	640	2,560
3	512	2,048
4	409.60	1,638.40
5	327.68	1,310.72
6	262.14	1,048.58
7	209.72	838.86
8	167.77	671.09
9	134.22	536.87
10	107.37	429.50

7.3.3. DB with Switchover to SL

Because the DB method never reaches a BV of zero, it is permissible to switch from this method to the SL method so that an asset's BVN will be zero.

This switchover occurs in the year in which an equal or a larger depreciation amount is obtained from the SL method

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7.3. The Classical Depreciation Methods

TABLE 7-1 The 200% DB Method with Switchover to the SL Method (Example 7-2)

		Depreciation Method		
Year, k	(1) Beginning-of-Year BV[a]	(2) 200% DB Method[b]	(3) SL Method[c]	(4) Depreciation Amount Selected[d]
1	$4,000.00	$800.00	>$400.00	$800.00
2	3,200.00	640.00	>355.56	640.00
3	2,560.00	512.00	>320.00	512.00
4	2,048.00	409.60	>292.57	409.60
5	1,638.40	327.68	>273.07	327.68
6	1,310.72	262.14	=262.14	262.14 (switch)
7	1,048.58	209.72	<262.14	262.14
8	786.44	167.77	<262.14	262.14
9	524.30	134.22	<262.14	262.14
10	262.16	107.37	<262.14	262.14
		$3,570.50		$4,000.00

[a] Column 1 for year k less column 4 for year k equals the entry in column 1 for year $k + 1$.
[b] 200%(= 2/10) of column 1.
[c] Column 1 minus estimated SV_N divided by the remaining years from the beginning of the year through the 10th year.
[d] Select the larger amount in column 2 or column 3.

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7.3.4. Units-of-Production Method

The *units-of-production* method can be used when the decrease in value of the assset is mostly a function of use, instead of time. The cost basis is allocated equally over the number of units produced over the asset's life. The depreciation per unit of production is found from the formula below.

$$\frac{B - SV_N}{(\text{Estimated lifetime production units})}$$

Chapter 7
7.3. The Classical Depreciation Methods

EXAMPLE 7-3 **Depreciation Based on Activity**

A piece of equipment used in a business has a basis of $50,000 and is expected to have a $10,000 SV when replaced after 30,000 hours of use. Find its depreciation rate per hour of use, and find its BV after 10,000 hours of operation.

Solution

$$\text{Depreciation per unit of production} = \frac{\$50{,}000 - \$10{,}000}{30{,}000 \text{ hours}} = \$1.33 \text{ per hour.}$$

$$\text{After 10,000 hours, BV} = \$50{,}000 - \frac{\$1.33}{\text{hour}}(10{,}000 \text{ hours}), \text{ or BV} = \$36{,}700.$$

Chapter 7
7.6. Introduction to Income Taxes

There are many different types of taxes.

- Income taxes are assessed as a function of gross revenues minus allowable expenses.
- Property taxes are assessed as a function of the value of property owned.
- Sales taxes are assessed on the basis of purchase of goods or services.
- Excise taxes are federal taxes assessed as a function of the sale of certain goods or services often considered nonnecessities.

We will focus on income taxes.

Chapter 7
7.6. Introduction to Income Taxes

Taking taxes into account changes our expectations of returns on projects, so our MARR (after-tax) is lower.

$$\text{Before-tax MARR} \approx \frac{\text{After-tax MARR}}{\text{1-effective income tax rate}}$$

The after-tax MARR should be *at least* the tax-adjusted weighted average cost of capital (WACC).

$$WACC = \lambda(1-t)i_b + (1-\lambda)e_a$$

λ = fraction of a firm's pool of capital borrowed from lenders
t = effective income tax rate as a decimal
i_b = before-tax interest paid on borrowed capital
e_a = after-tax cost of equity capital

Depreciation is not a cash flow, but it affects a corporation's taxable income, and therefore the taxes a corporation pays.

Taxable income = gross income - all expenses except capital invest – depreciation deductions.

Chapter 7
7.6. Introduction to Income Taxes

Pause and solve

Last year Acme, Inc. had $16.4 million in revenue, $1.2 million of operating expenses, and depreciation expenses of $5.4 million. Using the corporate federal tax rates from the table provided in the text, what is the approximate federal tax this corporation will have to pay for this tax year?

Solution

Last year Acme, Inc. had $16.4 million in revenue, $1.2 million of operating expenses, and depreciation expenses of $5.4 million. Using the corporate federal tax rates from the table provided in the text, what is the approximate federal tax this corporation will have to pay for this tax year?

$$\text{taxable income} = \$16.4 - \$1.2 - \$5.4 = \$9.8 \text{ million}$$

$$\text{tax} = \$113,900 + (\$9,800,000 - \$335,000) * 0.34$$

$$\text{tax} = \$3,332,000$$

Chapter 7
7.6. Introduction to Income Taxes

EXAMPLE 7-8 **Determination of Taxable Income**

A company generates $1,500,000 of gross income during its tax year and incurs operating expenses of $800,000. Property taxes on business assets amount to $48,000. The total depreciation deductions for the tax year equal $114,000. What is the taxable income of this firm?

Solution

Based on Equation (7-14), this company's taxable income for the tax year would be

$$\$1{,}500{,}000 - \$800{,}000 - \$48{,}000 - \$114{,}000 = \$538{,}000.$$

Chapter 7
7.6. Introduction to Income Taxes

EXAMPLE 7-9 **Calculating Income Taxes**



Suppose that a firm for a tax year has a gross income of $5,270,000, expenses (excluding capital) of $2,927,500, and depreciation deductions of $1,874,300. What would be its taxable income and federal income tax for the tax year, based on Equation (7-14) and Table 7-5?

Solution

$$\begin{aligned}
\text{Taxable income} &= \text{Gross income} - \text{Expenses} - \text{Depreciation deductions} \\
&= \$5{,}270{,}000 - \$2{,}927{,}500 - \$1{,}874{,}300 \\
&= \$468{,}200
\end{aligned}$$

Income tax =	15% of first $50,000	$7,500
	+ 25% of the next $25,000	6,250
	+ 34% of the next $25,000	8,500
	+ 39% of the next $235,000	91,650
	+ 34% of the next $133,200	45,288
	Total	$159,188

The total tax liability in this case is $159,188. As an added note, we could have used a flat rate of 34% in this example because the federal weighted average tax rate at taxable income = $335,000 is 34%. The remaining $133,200 of taxable income above this amount is in a 34% tax bracket (Table 7-5). So we have 0.34($468,200) = $159,188.

Chapter 7
7.8. Gain (Loss) on the Disposal of an Asset

The disposal of a depreciable asset can result in a gain or loss based on the sale price (market value) and the current book value

$$[\text{gain (loss) on disposal}]_N = MV_N - BV_N$$

A gain is often referred to as *depreciation recapture*, and it is generally taxed as the same as ordinary income. A loss is a *capital loss*. An asset sold for more than it's cost basis results in a *capital gain*.

Chapter 7
7.8. Gain (Loss) on the Disposal of an Asset

Pause and solve

Acme Casting and Molding sold a piece of equipment during the current tax year for $67,000. This equipment had a cost basis of $210,000 and the accumulated depreciation was $153,000. Assume the effective income tax rate is 40%. Based on this information, what is

a. the gain (loss) on disposal,
b. the tax liability (or credit) resulting from this sale, and
c. the tax liability (or credit) if the accumulated depreciation was $125,000 instead of $153,000?

Solution

a. [gain(loss)] = $67,000 – ($210,000 - $153,000)
= $10,000 gain

b. tax liability = $10,000(0.4) = $ 4,000

c. loss = $ 67,000 - $ 85,000 = - $ 18,000
tax credit = $ 18,000(0.4) = $7,200

Chapter 7
7.8. Gain (Loss) on the Disposal of an Asset

EXAMPLE 7-11 **Tax Consequences of Selling an Asset**



A corporation sold a piece of equipment during the current tax year for \$78,600. The accounting records show that its cost basis, B, is \$190,000 and the accumulated depreciation is \$139,200. Assume that the effective income tax rate as a decimal is 0.40 (40%). Based on this information, what is

(a) the gain (loss) on disposal,

(b) the tax liability (or credit) resulting from this sale,

(c) the tax liability (or credit) if the accumulated depreciation was \$92,400 instead of \$139,200?

Solution

(a) The BV at the time of sale is \$190,000 − \$139,200 = \$50,800. Therefore, the gain on disposal is \$78,600 − \$50,800 = \$27,800.

(b) The tax owed on this gain is −0.40(\$27,800) = −\$11,120.

(c) With $d_k^* = \$92{,}400$, the BV at the time of sale is \$190,000 − \$92,400 = \$97,600. The loss is \$78,600 − \$97,600 = −\$19,000. The tax credit resulting from this loss on disposal is −0.40(−\$19,000) = \$7,600.